Lead Creation Workflow

Step 3 & Step 4 — Design & Implementation Guide

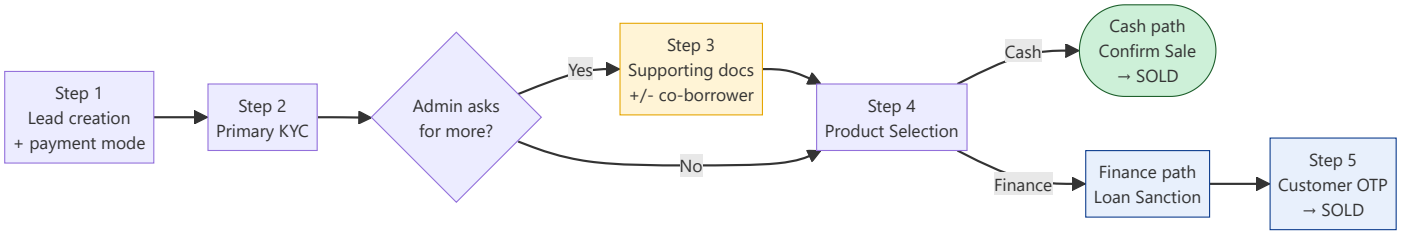
iTarang — Dealer Management CRM

Prepared for internal engineering review

Source of truth: `docs/superpowers/plans/lead creationworkflow.md`

1. Where Step 3 & Step 4 fit in the full lead flow

Step 3 is a **conditional** re-verification loop, shown only when the admin needs more from the dealer after Step 2. Step 4 is where **payment mode (Cash vs Finance) splits the workflow** — Finance routes through an admin loan sanction and a Step 5 OTP confirmation; Cash is a one-shot dealer-authorised sale.



> **Rule:** Step 4 and Step 5 are *blocked* until Step 3 is cleared by admin — unless Step 3 was never triggered for this lead, in which case Step 2 → Step 4 directly.

2. Step 3 — Conditional Re-verification

2.1 Purpose

After Step 2, admin may need (a) more supporting documents from the primary applicant, and/or (b) a co-borrower if the applicant's financials are weak (CIBIL < 650, insufficient income, high DTI). Step 3 handles both without restarting the flow.

2.2 When Step 3 appears

Only when `lead.kyc_status` is one of:

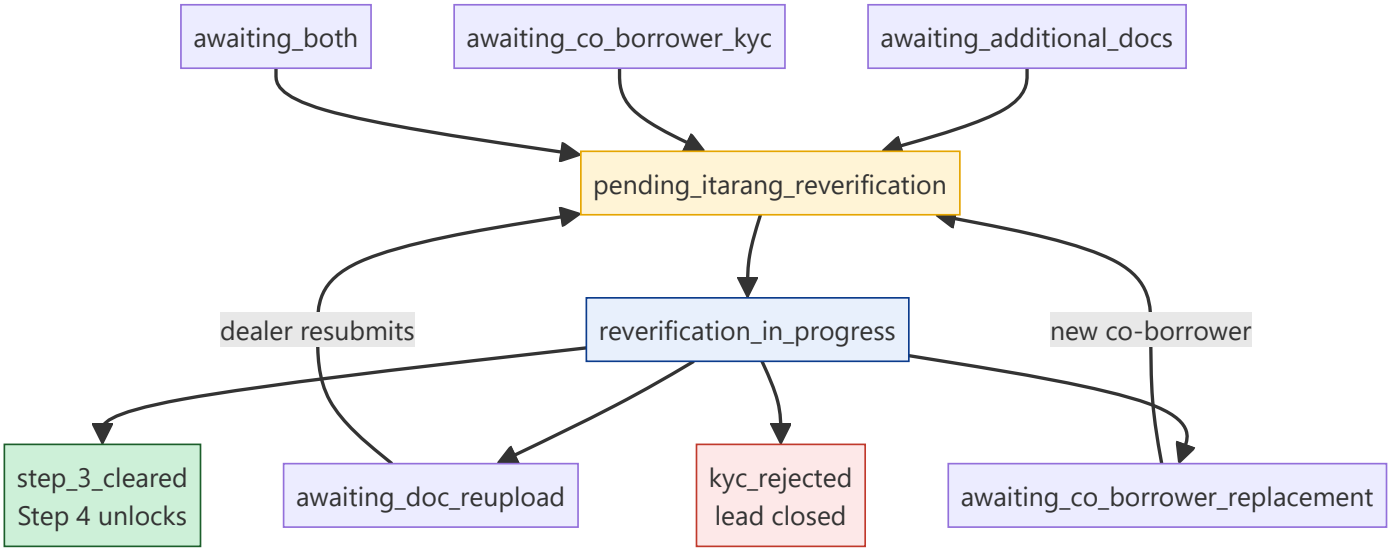
- `awaiting_additional_docs` — dealer uploads more supporting docs
- `awaiting_co_borrower_kyc` — dealer does full co-borrower KYC
- `awaiting_both` — both sections

Otherwise the progress bar jumps Step 2 → Step 4 directly.

2.3 Three admin-request scenarios

Scenario	Trigger	Dealer screen shows
1. Supporting docs only	PAN unclear, bank stmt incomplete, RC illegible, etc.	Section A only
2. Co-borrower only	CIBIL < 650, income low, high DTI, no credit history	Section B only
3. Both	Both problems simultaneously	Section A + Section B

2.4 Step 3 status lifecycle



2.5 Dealer-side Step 3 screen — sections

Section	Shown when	Contains
Admin Request Banner	Always	Read-only: admin's request reason + deadline
Section A — Supporting Docs	`awaiting_additional_docs` or `awaiting_both`	Dynamic upload cards — one per doc admin requested. No static list. Each card: name, admin reason, upload button, status badge (Pending Review / Verified / Rejected).
Section B — Co-Borrower KYC	`awaiting_co_borrower_kyc` or `awaiting_both`	9-field profile form + 11 document cards + consent block (digital eSign via DigiO, or manual upload) + coupon validation.
Submit bar	Always (bottom)	Back · Save Draft · Preview Profile · Submit for Verification.

> **Submit gate:** all required docs uploaded + (if co-borrower) form complete + consent captured + coupon validated. Supporting-docs-only case does **not** require a new coupon — the original lead coupon already covers it.

2.6 Admin-side Step 3 — appended review panels

Admin opens the existing Step 2 KYC screen and finds new panels appended below (primary KYC stays read-only):

#	Panel	When shown	Per-card actions
1	Primary KYC (existing)	Always	Read-only
2	Supporting Docs (appended)	If docs requested	Approve · Reject · Request More Docs (per doc card, mutually exclusive)
3	Co-Borrower KYC (appended)	If co-borrower requested	Same 3-button action per verification card (Aadhaar, PAN, Face Match, Bank, Address, CIBIL)
4	Final Decision	Always (pinned bottom)	Approve Lead · Reject Lead · Dealer Action Required

2.7 Final Decision Panel — gating rules

Card state combo	Approve Lead	Dealer Action Required	Reject Lead
All cards green (Approved)	✅ Enabled	Available	Always available
Any card Rejected or Request Docs	🔒 Disabled	✅ Expected action	Always available

Co-borrower replacement: If CIBIL or identity fails on the co-borrower, admin uses *Dealer Action Required* → status becomes `awaiting_co_borrower_replacement`. The `co_borrower_requests.attempt_number` increments so audit logs track how many co-borrowers were tried.

2.8 Step 3 — DB tables (schema changes needed)

Table	Status	Purpose
`additional_document_requests`	Build	One row per admin request — request_type, documents_requested[], reason, deadline, status
`additional_document_uploads`	Build	One row per uploaded file against a request
`co_borrower_requests`	Build	One row per co-borrower request — includes `attempt_number`
`co_borrowers`	Exists	Full co-borrower profile
`co_borrower_documents`	Exists	11 doc slots per co-borrower
`co_borrower_consent`	Partial	Digital eSign / manual upload tracking; consent_type NULL until picked

2.9 Step 3 — API endpoints

Method + Path	Who	Effect
`POST /api/admin/lead/:id/request-additional-docs`	Admin	Triggers Scenario 1; writes `awaiting_additional_docs`
`POST /api/admin/lead/:id/request-co-borrower`	Admin	Triggers Scenario 2/3
`POST /api/lead/:id/step-3/upload-document`	Dealer	Uploads one supporting doc against the request
`POST /api/lead/:id/step-3/co-borrower`	Dealer	Submits co-borrower form + docs
`POST /api/lead/:id/step-3/submit`	Dealer	Final submit → `pending_itarang_reverification`
`POST /api/admin/lead/:id/step-3-decision`	Admin	Per-card Approve/Reject/Request
`POST /api/admin/lead/:id/step-3-final-decision`	Admin	Approve Lead / Reject Lead / Dealer Action Required

3. Step 4 — Product Selection (the Cash / Finance split)

3.1 Purpose

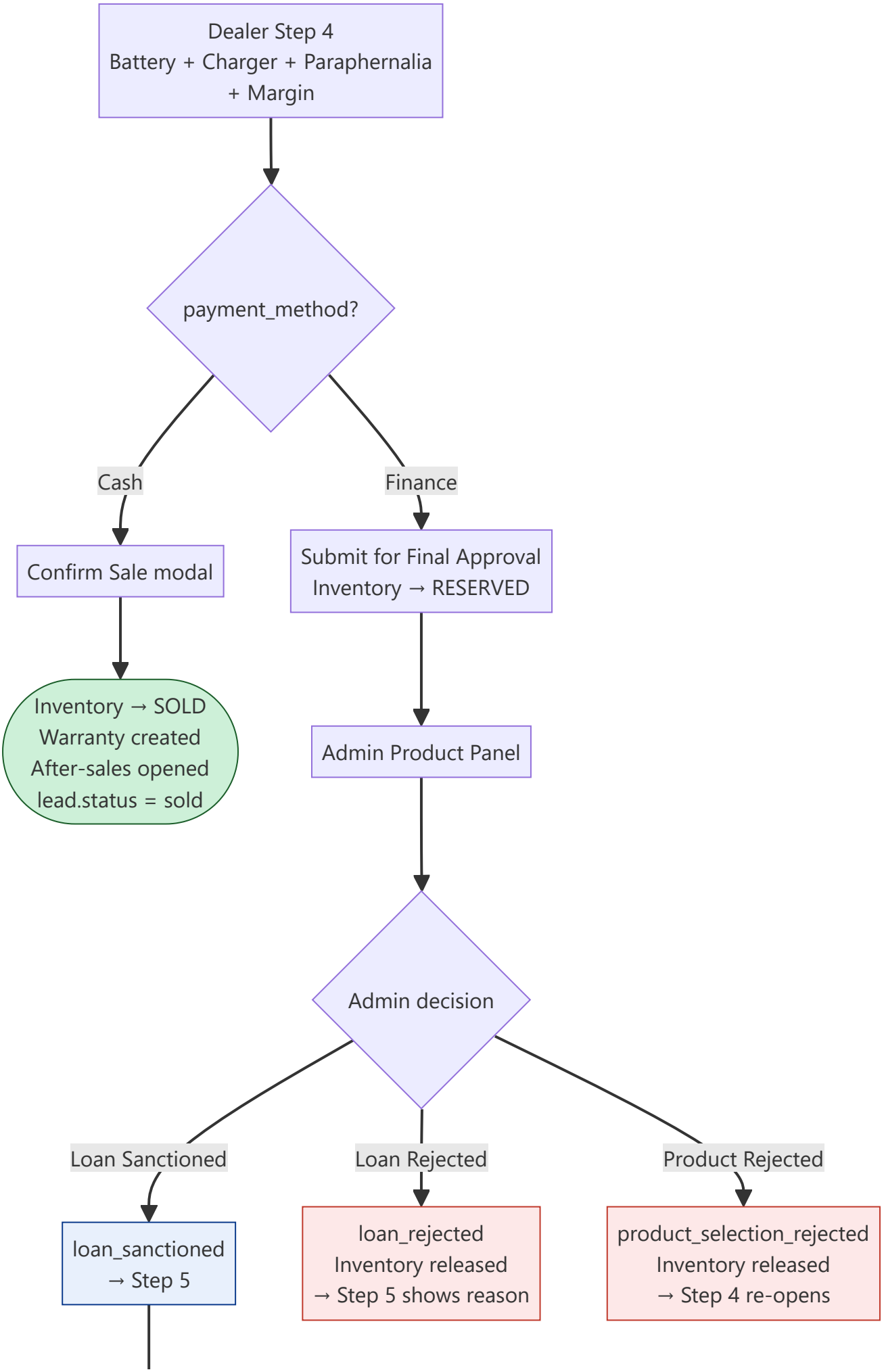
Dealer maps the physical product — battery + charger + paraphernalia — to this lead, sets their margin, and submits. **This is where the workflow diverges based on payment mode.**

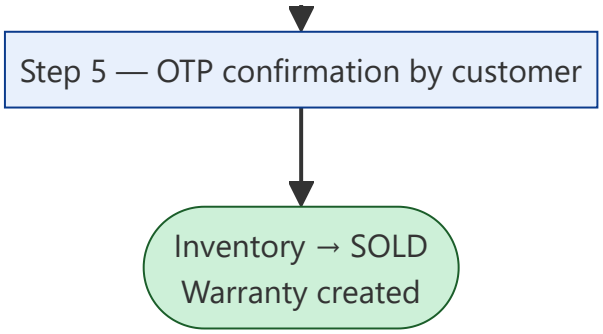
3.2 Entry gate

Payment mode	Entry condition	Admin step after?
Finance / Dealer Finance	`kyc_status = step_3_cleared` or `kyc_approved`	**Yes** — admin loan sanction
Cash	`payment_method = 'Cash'`	**No** — dealer confirms sale directly
Anything else	—	🔒 Blocked; redirect to last valid step

Gated by `GET /api/lead/:id/step-4-access` Exists .

3.3 Branching diagram — Cash vs Finance at submit


Dealer Step 4
Battery + Charger + Paraphernalia
+ Margin
payment_method?
Cash
Finance
Confirm Sale modal
Submit for Final Approval
Inventory → RESERVED
Inventory → SOLD
Warranty created
After-sales opened
lead.status = sold
Admin Product Panel
Admin decision
Loan Sanctioned
Loan Rejected
Product Rejected
loan_sanctioned
→ Step 5
loan_rejected
Inventory released
→ Step 5 shows reason
product_selection_rejected
Inventory released
→ Step 4 re-opens



3.4 Cash vs Finance — side-by-side

Aspect	Finance path	Cash path
Submit button	"Submit for Final Approval"	"Confirm Sale"
Confirmation modal	—	Required, with full summary
Inventory on submit	`reserved`	`sold` — immediate, skips reserved
Admin queue	Yes — admin sanctions loan	None — dealer is sole authorising party
Warranty	Created only after Step 5 OTP	Created immediately on confirm
After-sales record	Opens after Step 5	Opens immediately on confirm
Final `lead.status`	`sold` (via Step 5)	`sold` (one shot)

3.5 Dealer-side Step 4 screen — 6 sections

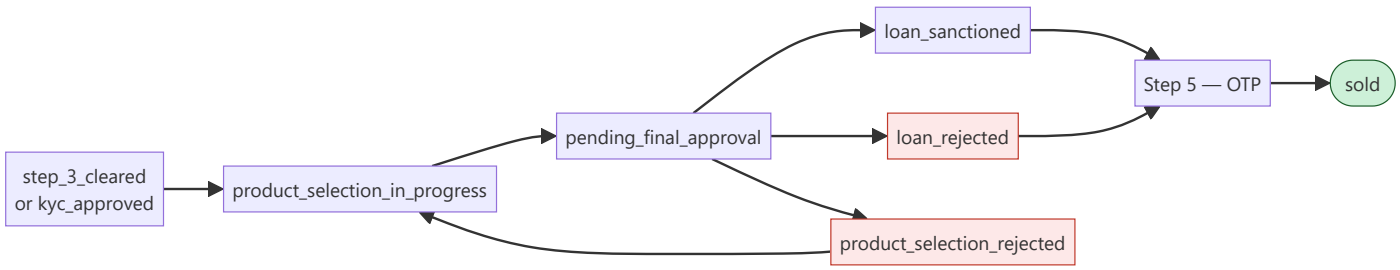
Section	Content	Key rule
A. Category + Sub-category	Pre-filled from Step 1; editable	Changing category clears prior battery/charger
B. Battery Selection	Filtered by category. Fields: serial, model, invoice date, age, SOC %, status	**Sorted oldest-first.** Ageing badges: 0-90 normal · 91-180 orange (Ageing Stock) · >180 red (Old Stock — Prioritise). Oldest unit flagged "Recommended".
C. Charger Selection	Filtered by battery-model compatibility	Same ageing rules
D. Paraphernalia	Digital SOC, Volt SOC, Harness variant, other accessories	Count-based, not serial-tracked. Checked against dealer stock at submit.
E. Pricing & Margin	Base prices system-controlled (read-only). Dealer enters margin.	Final price auto-calculated.
F. Submit	Button label + behaviour split by payment_mode (see 3.4)	Button gated on: battery selected (available + correct category), charger selected (available + compatible)

3.6 Admin-side — Product Selection Panel (Finance leads only)

Appended below the KYC screen, following the same append pattern as Step 3. Cash leads never enter this panel.

Action	Lead status after	Inventory impact
Loan Sanctioned (opens 10-field loan form)	`loan_sanctioned`	Stays `reserved`. Routes to Step 5.
Loan Rejected (mandatory reason)	`loan_rejected`	`reserved → available`. Routes to Step 5 showing reason.
Product Rejected	`product_selection_rejected`	`reserved → available`. Step 4 re-opens on dealer side.
Download Profile	no change	Streams ZIP: summary PDF + all KYC + supporting docs + co-borrower docs + product summary.

3.7 Step 4 — Finance path state machine



3.8 Step 4 — DB tables

Table	Status	Purpose
`product_selections`	Partial	One row per lead — battery/charger serial, paraphernalia JSON, margin, final price, admin_decision
`loan_sanctions`	Partial	One row per sanctioned loan — 10 loan fields + sanctioned_by/at
`loan_rejections`	Partial	Rejection reason + rejected_by/at
`warranties`	Build	warranty_id, start_date, battery serial
`after_sales_records`	Build	Service lifecycle tracking
`inventory`	Exists	status transitions: `available → reserved → sold` (Finance) or `available → sold` (Cash)

3.9 Step 4 — API endpoints (current state)

Endpoint	Status
`GET /api/lead/:id/step-4-access`	Exists
`GET /api/inventory/dealer/:dealerId/batteries`	Exists
`GET /api/inventory/dealer/:dealerId/chargers`	Exists
`GET /api/inventory/dealer/:dealerId/paraphernalia`	Exists
`POST /api/lead/:id/submit-product-selection`	Exists (Finance)
`POST /api/lead/:id/confirm-cash-sale`	Exists (Cash)
`POST /api/admin/lead/:id/sanction-loan`	Exists
`POST /api/admin/lead/:id/reject-loan`	Exists
`POST /api/admin/lead/:id/product-selection`	Exists
`GET /api/admin/lead/:id/download-profile`	Exists

4. Implementation status & build order

4.1 What's already scaffolded in the codebase

- Dealer pages: `dealer-portal/leads/[id]/kyc/interim/` (Step 3), `product-selection/` (Step 4), `step-5/`
- Admin page: `admin/product-review/[leadId]`
- Inventory APIs (batteries, chargers, paraphernalia)
- Step 4 submit + cash confirm + loan sanction / reject / product-selection APIs
- Schema: `co_borrowers`, `co_borrower_documents`

4.2 Step 3 — what's still to build

- `additional_document_requests` + `additional_document_uploads` tables
- `co_borrower_requests` table (with `attempt_number`)
- Admin "Request More Docs" and "Request Co-Borrower" modals + POST endpoints
- Per-card 3-button action (Approve / Reject / Request More Docs) on the admin side
- Final Decision Panel — the gating logic (Approve enabled only when all cards green)
- Dealer-side dynamic rendering of Section A vs Section B vs both
- Co-borrower consent flow: digital eSign (reuses existing DigiO consent infra) + manual upload
- Notifications (SMS + email + dashboard push) for the 6 Step 3 events

4.3 Step 4 — what's still to polish

- Inventory ageing badges (0-90 / 91-180 / >180) on the UI
- IoT SOC fetch + stale-data indicator
- Race-condition guard on submit (two dealers picking the same serial)
- Admin "Product Rejected" action wiring (Loan Sanctioned / Rejected already done)
- Warranty + after-sales record creation on cash confirm
- ZIP download assembly for the profile (schema wired, assembly is the work)

4.4 Suggested build order

1. **Step 3 tables first** — unblocks both dealer and admin sides.
2. **Admin "Request More Docs" modal + API** — the trigger that opens Step 3.
3. **Dealer Step 3 dynamic rendering** — Section A + Section B driven by backend-returned request data.
4. **Admin Step 3 append panels + 3-button per-card logic** — largest UI chunk.
5. **Final Decision Panel gating logic** — Approve only when all cards green.
6. **Co-borrower consent flow** — digital first (reuses DigiO), manual is simpler.
7. **Step 4 polish** — ageing badges, IoT SOC, race-condition guard.
8. **Product Rejected action + Warranty + After-sales creation** — finance & cash loose ends.
9. **Notifications + ZIP download** — cross-cutting, can be parallelised.

5. Anticipated team questions

> **Q: Why is Step 3 conditional?**
> Most leads (~70% per the BRD) pass primary KYC cleanly. Forcing every dealer through Step 3 adds friction and slows throughput. The conditional design keeps the flow short for clean applicants and targets the re-verification loop only when admin genuinely needs more from the dealer.

> **Q: What happens if admin changes their mind after requesting?**
> Admin can still use "Approve Lead" from the Final Decision Panel if the cards subsequently turn green, or issue a fresh request. There is no "undo" of the original request — it's an audit trail.

> **Q: Why is the cash flow one-shot with no admin step?**
> In cash the dealer is the sole authorising party. There is no loan to sanction and no lender to align. Forcing admin approval would only slow inventory turns and revenue. Admin still sees the sale retrospectively via the Download Profile action.

> **Q: How do we prevent the same battery serial being picked by two dealers simultaneously?**
> Server-side race-condition guard on `/submit-product-selection` and `/confirm-cash-sale` : a transactional `SELECT ... FOR UPDATE` + status check before transitioning `inventory.status` . If another request already transitioned it, return a clear "serial no longer available — please re-select" error.

6. References

- Full BRD: `docs/superpowers/plans/lead creationworkflow.md` — Step 3 at lines 3311–3649, Step 4 at lines 3651–4099
- Existing code: `src/app/(dashboard)/dealer-portal/leads/[id]/` , `src/app/api/admin/lead/[id]/` , `src/app/api/inventory/`
- Schema: `src/lib/db/schema.ts`